SAT–Terrestrial Coexistence With Adaptive Beam and Power Control: An SDR-Based Emulation

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***Abstract*—With increasing demand for spectrum, the 12 GHz band is ideal for future 5G deployments, due to its abundant bandwidth and favorable propagation characteristics. Despite these benefits, aspiring 5G terrestrial networks pose critical interference and coexistence challenges for incumbent, in-band, and susceptible satellite (SAT) Digital Broadcasting Stations (DVB-S2). This paper leverages directional beam control and power adaptation through a spectrum sharing framework, minimizing interference on terrestrial SAT receivers. Opposing traditionally rigid spectrum sharing regulations, our framework implements an efficient and reproducible Joint Q-Learning algorithm through a Software-Defined Radio (SDR) emulation on the COSMOS testbed. Evaluating this model against realistic 5G transmission parameters and ETSI Rural Macro case study, a notable highlight of our method is demonstrated through the DVB-S2 Receiver Packet Error Rate (PER), restoring up to 98% of the DVB-S2 link's ideal performance through coordinated Radio Resource Management (RRM). Furthermore, our model sustains a maximum of 48% standalone 5G network capacity in the presence of SAT downlink, showcasing acceptable coexistence despite simultaneous band usage.**

***Index Terms*—Spectrum Sharing, Power Control, Beam Control, 5G, DVB-S2, Satellite Systems**

I. Introduction

5G New Radio (NR) and voracious spectrum use cases have promoted major standards across cellular technology with ultra low latency, gigabit-speed connectivity and enhanced reliability. Addressing these requirements, commercial 5G deployments have explored two major frequency bands: FR1 (less than 7.125 GHz) and FR2 (24-52.6 GHz) [1]. While FR1 suffers from limited spectrum accommodation, FR2 band provides uninterruptible 800 MHZ bandwidth at the cost of ineffective penetration ability in macro-cellular deployments. In this context, the FR3 band (7.125-24 GHz) has garnered attention by offering bandwidth availability and base station (BS) coverages 2.33 times greater than FR2 counterparts [2]. However, in the U.S., this band is used by incumbent users from Satellite (SAT) Digital Video Broadcasting Satellite Service (DVB-S2), which operate under a combined regulatory framework for mutual coexistence [3]. Thus, an interference analysis and spectrum sharing framework for incumbent users is imperative before 5G deployment in FR3.

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Studying SAT-5G coexistence in FR3 bands requires understanding of the FR3 environment's idiosyncrasies and emulation of waveform path loss. Existing work on SAT-5G coexistence has considered a simulation based approach through atmospheric and path loss modeling [4], taking into account in depth constraints like 50% User Equipment (UE) loading. Measurements in FR3 (e.g. 16.95 GHz) have been carried out in [5], where they validate their FR3 modeling through robust measurements and angular models, comparing their results to theorized 3GPP models [6]. In this paper, we consider an emulation based approach using a mix of simulation and measurements on the COSMOS [7] Testbed to study machine learning based coexistence between DVB-S2 and 5G at 12 GHz. Specifically, we consider a realistic set of path losses and directivity analysis during the baseband processing of DVB-S2 waveforms in our study.

Given this background, earlier works have explored spectrum coexistence through spectrum sensing, sharing and RRM with emphasis on machine learning based approaches. In recent work, a deep learning version of energy detection was deployed in [8] for the SPN-43 radar detection, collecting 14,000 hand-labeled spectrograms for their CNN/LSTM classification. Given their extensive training data, this approach to spectrum sensing requires large datasets and static sensor embeddings; whereas, a collaborative, real-time feedback loop as considered in our work, to reach agreement on spectrum usage can substantially reduce computational overhead. From [9], a supervised model called DeepAlloc was introduced, inferring spectrum allocation with large volumes of synthetic data; however, DeepAlloc's limited capability for dynamic scenarios, varying 5G gain, and unpredictable directionality makes refinement difficult and nonadaptive to generalized FR3 environments. This highlights the necessity of an adaptive coexistence by directly handling primary and secondary user inputs, ensuring scalability and adaptability without retraining on enormous datasets. In addition, RRM for satellite coexistence has produced optimization controls for several transmission parameters: power, beam, or Dynamic Exclusion Zones (DEZ). In [10], context-aware RRM policies were proposed based on DEZ through a simulated 5G testbed. While it provides validation for future FR3 experimentation, the simulated approach hinders scalability for hardware RRM management. Meanwhile, Zhang et al. [11] applied deep RL methods—A3C and DPPO—to optimize power control for secondary users in cognitive radio networks, relying on distributed sensors and power adaptation. However, their work does not address the complexities of communication software stacks, physical waveforms, and band accommodation. Similarly, the authors in [12] implement a channel allocation-based power control policy to manage BS resources but underused 5G BS's beam adaptivity capabilities. Relatedly, our earlier work [13] studied 5G-SAT coexistence by employing an omnidirectional OFDM-based 5G model with LQR-based power/zone control, but lacked adaptive beamsteering, standards-compliant 5G NR integration, and adaptability to robust interference due to rigid nature of the LQR controller. These shortcomings necessitate a hardware-based RRM solution for shared-spectrum environments that can dynamically adapt power, beamforming, and exclusion zones, while also being aware to the unique propagation characteristics of FR3 band waveforms.

Specifically, this paper establishes an effective model with Joint Power and Beam Control RRM, providing controllable coexistence scenarios in the FR3 band. With data scarcity of SAT-5G coexistence modeling, our framework bolsters comprehension and control of the 5G base station (BS) transmitter through a Software-Defined Radio (SDR) implementation, while managing interference to the DVB-S2 downlink. Through reproducible and realistic waveform emulation, our approach makes the following contributions:

1) Emulate a real-time SAT–5G coexistence scenario with full Amarisoft 5G NR protocol stack and ETSI standards-compliant DVB-S2 SAT waveforms on COTS SDRs.
2) Incorporate a Joint Q-Learning Power and Directivity Controller through a Vivaldi antenna setup, minimizing terrestrial interference at the SAT receiver and maximizing 5G throughput.
3) Assessing the system model performance through a ETSI Rural Macro case study emulated on the COSMOS testbed.

The rest of the paper is organized as follows: Section II explains the Coexistence Model, followed by the Emulation Framework in Section III, Section IV details the Rural Macro interference scenario, and Section V explains the RRM techniques with emulated results. Section VI concludes with future directions.

II. Spectrum Coexistence Model

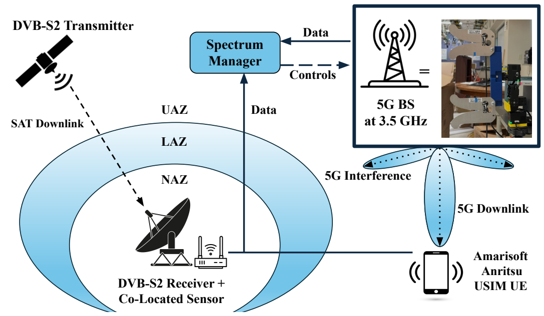


Fig. 1: Spectrum Coexistence Model

As shown in Fig. 1, the spectrum coexistence model considers co-located 5G and DVB-S2 downlinks operating in a shared frequency band and assigned to a spectrum manager. The spectrum manager framework intakes direct data feedback from the 5G BS–UE link and a DVB-S2 SAT transmitter–receiver link, applying real-time RMM control updates on the 5G BS. The 5G BS incorporates a dual Vivaldi antenna setup to control transmission through beam steering on azimuth angle θ and regulating transmission gain.

The 5G BS and UE continuously reports link quality metrics—Signal-to-Interference-plus-Noise Ratio (SINR) in dB, Modulation and Coding Scheme (MCS), Reference Signal Received Power (RSRP) in dBm, and bitrate in Mbps to the spectrum manager. Simultaneously, the DVB-S2 Receiver and Co-Located Sensor provide Packet Error Rate (PER) and SNR updates, relaying co-channel interference information from the terrestrial 5G transmissions in the Unlimited, Limited, or No Access Zones (UAZ, LAZ, NAZ). By processing data and updating the 5G beam away from the SAT receiver's line-of-sight, the model limits emitted power during high-interference periods and retains a substantial SAT link integrity. Due to real-time control and connected data piping, the spectrum manager can issue control commands to the 5G BS, while scaling for more RRM inputs if there is a requirement for data-driven, intensive, or larger 5G networks.

Through coordinated power–direction control and multi-source feedback, the system achieves interference-aware coexistence between terrestrial 5G and DVB-S2 downlink communications. Thus, the framework prevents the interruption of the DVB-S2 Receiver Phased-Locked Loop, avoiding interference scenarios where the frequency offset could deter SAT link quality. Although this implementation currently considers only the 5G BS downlink, the emulation framework can be extended (and scaled up)—without loss of generality—to include 5G uplink transmissions, enabling broader interference management in bidirectional 5G communication scenarios.

III. Emulation Framework

A. Hardware and Software Setup

The experimental setup is implemented on COSMOS Testbed, executing RRM strategies and physical transmissions with X310 USRP devices. The testbed is configured for heterogeneous network emulation, incorporating terrestrial 5G systems alongside a SAT link. Each node is 3 ft. apart. Directionality of the 5G transmit antenna utilizes an azimuth motor connected to an Intel NUC for precision. A GNU Radio 3.10 environment on Linux Ubuntu 22.04 OSs supplements the network software and artificial characteristics of the DVB-S2 link, making the framework and experimentation reproducible for other hardware configuration. Furthermore, all data inputs and control updates are facilitated by the COSMOS Testbed's internal links between computers, as exemplified in Fig. 2, where the SAT RX and 5G UE are at azimuth angle $45°$ and $90°$ with respect to the 5G TX, respectively.

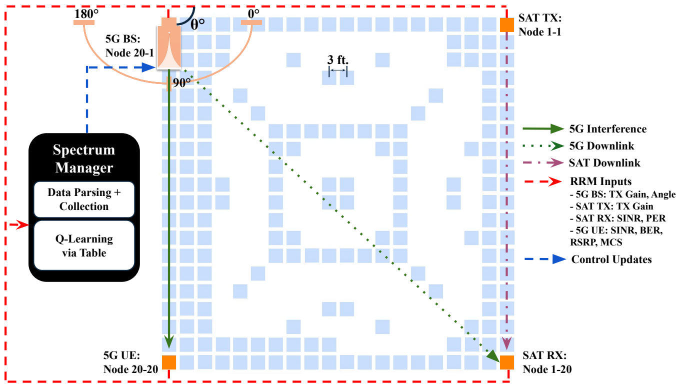


Fig. 2: Cosmos Experiment Layout

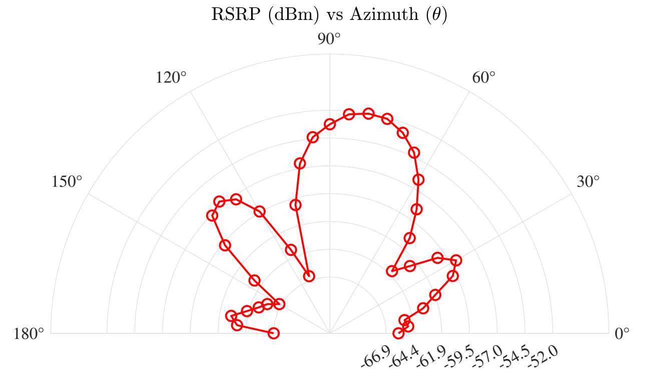


Fig. 3: Beam Pattern for Dual Vivaldi Antenna Setup at 3.5 GHz

B. Amarisoft 5G Waveform

We utilize Amarisoft for capable simulation of a 5G network, enabling an operational gNB and scalable set of UE. The software stack enables realistic 5G interference scenarios, validating our approach for rigorous standards from commercial BS. Complying with 3GPP standards, the 5G transmission is configurable for operating band, bandwidth, and payload modulation, ensuring consistent measurements and diagnostics as in Table I. With QPSK header modulation and 256 QAM payload modulation, this setup can support high data rate transmission, resembling a practical 5G deployment scenario for experimental validation and performance analysis.

C. Digital Video Broadcasting Waveform

Our setup emulates DVB-S2 transmission through GNU Radio, offering flexible configuration supporting different kind of modulations, code rates, and forward error correction. Based on ETSI's standards, we implement DVB-S2 physical layer and time synchronization, along with frame processing. The signal is modulated at 8PSK on a LDPC code rate of $\frac{3}{5}$, producing baseband transmission in DVB-S2. A continuously looped MPEG transport stream (.ts file) is transmitted, emulating a practical video broadcast scenario.

D. Limitations

Through the Rural Macro model derived from [6], the emulation of the path losses found in the DVB-S2 SAT downlink and DVB-S2 Receiver can be characterized in the 12.2-12.7 GHz range. However, due to the lack of physical infrastructure to transmit at FR3, the 100% overlap FR3 interference scenario is handled by baseband processing, where attenuation at the DVB-S2 receiver will replicate the effects. Specifically, we generate the baseband signals used for the emulation by transmitting RF signals at 3.5 GHz, followed by subsequent down-conversion to baseband frequency by the USRP X310's UBX-160 daughterboard. Thereby, COSMOS Grid's radio mapping features will produce the scalable waveforms in the FR1 band, accounting for downlink placement in the COSMOS testbed and adjusting it for FR3 impact assessment.

Time(s)	Quality Parameters		Transport Statistics	
	SINR (dB)	RSRP (dBm)	MCS	Bitrate (Mbps)
1	32.9	-53.4	23.7	122
2	32.3	-53.3	23.6	122
3	33.2	-53.3	23.6	122
4	32.2	-53.3	23.6	122
5	31.8	-53.3	23.6	123

TABLE I: Amarisoft Trace Output (Downlink Metrics)

E. Power Control and Directivity

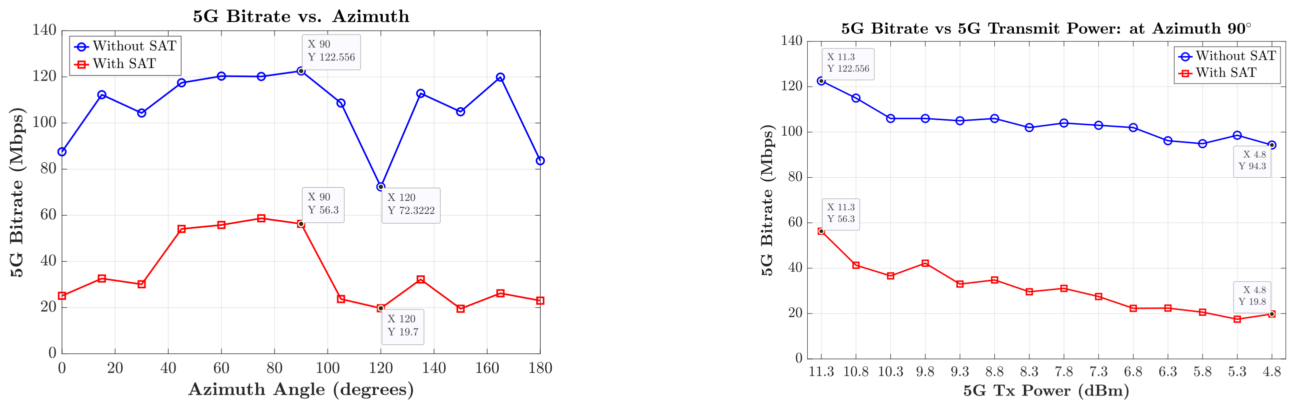
To evaluate the 5G gNB transmission power $\mathrm{P}_{5\mathrm{G}}^{\mathrm{TX}}$ impact on SAT-5G coexistence, a power control algorithm systematically varies the BS SDR gain $\mathrm{G}_{5\mathrm{G}}^{\mathrm{TX}} \in \{25, 25.5, \ldots, 31.5\}$ dB for fine-grained SAT link restoration. Azimuth angle θ is chosen by the Vivaldi antennas' spatial selectivity, directing away from the DVB-S2 Receiver and maximizing SAT received power $\mathrm{P}_{\mathrm{SAT}}^{\mathrm{RX}}$. Avoiding electromagnetic coupling at 3.5 GHz by spacing 2 UWB-4 antennas 8 cm. apart, interference is significantly reduced by identifying an ideal angle on 0-180° range and replicating beam steering of a 5G BS. The angle's corresponding RSRP as shown in Fig. 3 is utilized to calculate directivity, producing a standalone antenna BS gain $\mathrm{G}_{5\mathrm{G}}^{\mathrm{ANT}}$ (in dB) to reduce in-band interference and equate the 5G gNB transmit power to

$$\mathrm{P}_{5\mathrm{G}}^{\mathrm{TX}} = \mathrm{P}_{5\mathrm{G}}^{\mathrm{REF}} + \mathrm{G}_{5\mathrm{G}}^{\mathrm{TX}} + \mathrm{G}_{5\mathrm{G}}^{\mathrm{ANT}}(\theta), \tag{1}$$

where $\mathrm{P}_{5\mathrm{G}}^{\mathrm{REF}} = -20.2$ dBm is the X310 USRP reference power for the BS. Thus, $\mathrm{P}_{5\mathrm{G}}^{\mathrm{REF}} + \mathrm{G}_{5\mathrm{G}}^{\mathrm{TX}}$ accounts for power change in range of $\{4.8, 5.3, \ldots, 11.3\}$ dBm. Thereby, this enables power-performance tradeoffs for $\mathrm{P}_{5\mathrm{G}}^{\mathrm{TX}}$ and $\mathrm{P}_{\mathrm{SAT}}^{\mathrm{TX}}$, where reducing 5G transmit power corresponds to degradation of 5G throughput. Meanwhile, for SAT received power in dBm, it is strengthened with account to N noise floor, SAT and 5G path losses $\mathrm{L}_{\mathrm{SAT}}^{\mathrm{PATH}}$ and $\mathrm{L}_{5\mathrm{G}}^{\mathrm{PATH}}$, and SAT receiver gain $\mathrm{G}_{\mathrm{SAT}}^{\mathrm{ANT}}$ where the result becomes

$$\mathrm{P}_{\mathrm{SAT}}^{\mathrm{RX}} = \underbrace{(\mathrm{P}_{\mathrm{SAT}}^{\mathrm{TX}} - \mathrm{L}_{\mathrm{SAT}}^{\mathrm{PATH}})}_{\text{Desired SAT Signal}} + \underbrace{(\mathrm{P}_{5\mathrm{G}}^{\mathrm{TX}} - \mathrm{L}_{5\mathrm{G}}^{\mathrm{PATH}})}_{\text{5G Interference}} + \mathrm{G}_{\mathrm{SAT}}^{\mathrm{ANT}} + \mathrm{N}. \tag{2}$$

Ensuring experimental consistency and isolation of the 5G transmission power's effect, gains from the SAT transmitter, SAT receiver, and 5G receiver are constant through all measurements.



(a) 5G Bitrate vs. Azimuth Angle during SAT Scenarios w/ Constant Power

(b) 5G Bitrate vs. TX Power during SAT Scenarios w/ Constant Azimuth

Fig. 4: SAT-Induced degradation on 5G: Bitrate With and Without Satellite Interference

IV. Case Study: Rural Macro Region

A. Parameters

Repeating the Rural Macro scenario, the setup samples a 200 second DVB-S2 downlink pass for a SAT in geostationary orbit, while a 5G BS will have simultaneous transmission and reflect bandwidth occupancy of 5G as found in work [10]. Another experimentation will isolate both transmission links, establishing a control to examine FR3 impact and SAT-5G coexistence. Based on the ETSI study, a generalization for the UE connection distance will be 1 km away from the BS. The baseband processing will account for emulated distance between the SAT link and the BS link; thus, the ETSI Rural Macro with LOS path loss, Rician Fading, and parameters of Table II are demonstrated, with the 12.7 GHz FR3 adjustment frequency retained in channel modeling and path loss analysis.

B. Satellite-Induced Degradation in 5G Networks

Fig. 4(a) demonstrates the SAT transmission effects on the 5G bitrate and link performance for varying azimuth angles and differing DVB-S2 link presence during highest 5G power: 11.3 dBm. During DVB-S2 transmission, the 5G BS's performance achieves a minimum 19.7 Mbps and maximum 58.7 Mbps for a reliable link to the UE; however, regardless of the DVB-S2 presence, the emulated beamforming achieves discernible improvements between 45-99°, suggesting that directivity fundamentally preserves a modicum of 5G capacity. However, at the highest peak, the SAT degraded 5G bitrate becomes nearly 48% of the 5G standalone performance, ensuring sufficient performance for habitual 5G operations. As the highest peaks are also reflected without the presence

Experimental Properties	Units
FR1 Carrier Frequency	3.5 GHz
FR3 Adjustment Frequency	12.7 GHz
Sample Time Length	200 seconds
DVB-S2 Symbol Rate	10 Msamples/sec
Emulated Distance for DVB-S2 Link	35,786 km
5G BS Link Bandwidth	40 MHz
DVB-S2 Link Bandwidth	6 MHz
Standard Deviation Rician Fading	4 dB

TABLE II: Rural Macro Experiment Setup & Properties

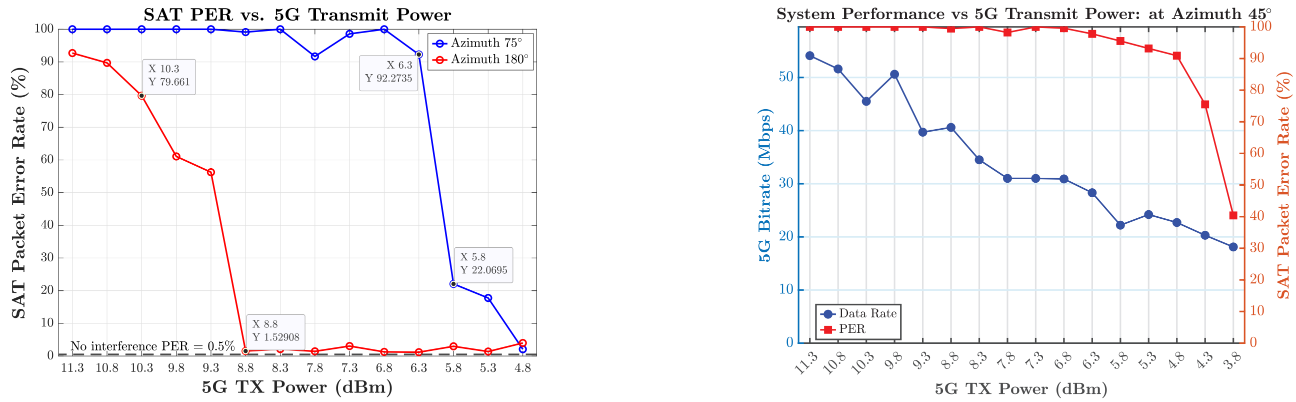
of the DVB-S2 link, this indicates that the Joint Power and Directivity algorithm's advantages are leveraging more directivity control for better performance.

Furthermore, as the relative shape of the bitrate curve is similar with or without the SAT link present, this verifies that the Vivaldi antenna setup and beamforming directly preserved a portion of the 5G BS link during FR3 coexistence scenarios. Therefore, despite how the SAT transmission interference can be prominent on the 5G BS link, the properties of the antennas and beamsteering can bolster the spatial distribution of throughput from the 5G BS.

Highlighted in Fig. 4(b), at a constant azimuth angle of 90° and depreciating transmit power, the 5G bitrate depreciates 2 times faster with the DVB-S2 link present than without. From nearly 60 Mbps to approximately 20 Mbps, the 5G link suffers significantly during SAT interference than without it, since it can maintain nearly 98 Mbps at its lowest transmit gain. Thus, this verifies substantial need for the 5G BS preservation by a power control algorithm, as it marginally improves 5G bitrate. It also indicates that preservation of the requisite signal fidelity of the 5G BS needs an incorporation of power adaptation during coexisting scenarios, bolstering the necessity and advantages of the Joint Power and Control algorithm. It is also observed that from both Fig. 4(a) and Fig. 4(b) that SAT disruption can greatly interfere with 5G demodulation and decoding, leading to the evident reduction in 5G throughput.

C. 5G-Induced Degradation in Satellite Systems

Fig. 5(a) illustrates the relationship between the SAT PER and the 5G transmit power at two azimuth angles: 75° and 180°. Regardless of the azimuth angle, the SAT PER shows a corresponding downward relationship, exhibiting a sharp drop over a narrow gain interval (typically within a 1–2 dB range) as the 5G transmit gain decreases. Acknowledging the effect in both angles, this is highly indicative of forward error correction (FEC) mechanisms in the DVB-S2 Receiver, where a thresholding effect substantiates an accumulated level bit errors to be corrected until interference exceeds the set limit. Thus, the generalized SAT performance maintains successful recovery of the original packets if the FEC can handle 5G BS



(a) SAT PER vs. TX Power at two different azimuth angles of the 5G BS.

(b) 5G Bitrate and SAT PER vs. TX Power w/ Constant Azimuth

Fig. 5: SAT PER/ 5G Bitrate vs. 5G Transmission (TX) Power in dBm for analyzing performance of Rural Macro Experiment.

transmit power at a certain threshold; thereby, this substantiates a feasible coexistence opportunity, where the 5G BS can be maintained at a power of 8.8 or 5.8 dBm as shown by the steep decline in PER. This threshold, in turn, translates to different radii of exclusion zone for varying azimuth angles.

Fig. 5(b) presents a dual-axis view of the system-level performance at azimuth 45°, where the 5G BS is directly aligned with the DVB-S2 Receiver This represents one of the worst-case interference scenarios. From the graph, the DVB-S2 link's PER remains above 90% for a huge proportion of 5G BS transmit power, dropping only after reduction past 4.3 dBm. As the 5G TX gain drops below this level, this reiterates that a directionality algorithm has high priority for coexisting management, ensuring the SAT performance especially.

V. Improving Spectrum Coexistence: Joint Radio Resource Management

A. *Joint RRM Framework*

The **Joint Power and Beam Control RRM** strategy employs a Q-Learning Controller for azimuth angle and transmit gain of the 5G BS. Specifically, this approach introduces reinforcement learning through a Q-table optimization policy, training and evaluating for real-time decisions on aggregated measurement data $\mathcal{D}$ and RRM inputs. Due to the Q-table's data, the action space either spans known azimuth angles θ in intervals of 15° or gain g through discretized 0.5 dB steps; however, if the observation determines the RRM state to be in a local optimum, the policy maximizes link performance by maintaining its current state. Incentivizing improvements to the coexistence scenario, a reward r_t improves aspired system performance through tunable weight scalars β and λ to obtain

$$r_t = \beta \cdot \mathrm{R}_t - \lambda \cdot \mathrm{PER}_t, \tag{3}$$

where R_t is the 5G bitrate and PER_t defines the PER of the DVB-S2 Receiver. To enable adaptability and stronger preferences to the disadvantaged network, β and λ can be modified toward an inclined result.

For a controllable yet swift policy, the controller updates Q-values for a given state s and action a, where the state corresponds to an temporally-approximated angle θ_t and gain g_t. Following the standard equation for Q-Learning as seen in Algorithm 1, the represented state-action pair is given a relative reward which contributes to the overall reward of the spectrum framework: total_reward. Through Learning Rate α and Discount Factor γ, the traversal of the Q-table policy can specify step size and priority on future rewards, respectively. Furthermore, the training policy follows an ϵ-greedy exploration strategy, where it selects the optimal action

$$a^* = \arg\max_a Q(s, a) \tag{4}$$

with probability $1 - \epsilon$, or a random action is selected with probability ϵ.

Algorithm 1: Q-Learning for Joint Power and Direction Control

Input: Dataset $\mathcal{D}$ with Azimuth, Gain, Bitrate, and PER
Output: Optimized Q-table and policy $\pi(a, g) = \arg\max_u Q[a, g, u]$

```
Define: θ ∈ {0°, 15°, ..., 180°}, g ∈ {25, 25.5, ..., 31.5}
Actions: a ∈ A = {increase_gain, decrease_gain,
    rotate_left, rotate_right, stay}
Initialize Q-table: Q(θ, g, a) ← 0, ∀θ, g, a
for episode ∈ [1, 2, ..., E] do
    s_0 = (θ_0, g_0) ∈ D ;    // Sample initial state
    t ← 0, total_reward ← 0
    while t < 100 do
        with prob ε: a_t ← random action
        else: a_t ← argmax_a Q(θ_t, g_t, a)
        if a_t = stay then
            s_{t+1} ← s_t
            r_{t+1} ← r_t + 0.1 ;    // Stability bonus
        else
            s_{t+1} ← simulate_action(s_t, a_t, D)
            r_{t+1} ← β · Bitrate − λ · PER
        Q(θ_t, g_t, a_t) ← (1 − α)Q(θ_t, g_t, a_t) + α[r_{t+1} +
            γ max_a Q(θ_{t+1}, g_{t+1}, a)]
        t ← t + 1
        total_reward ← total_reward + r_{t+1}
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B. *Joint RRM Control Case Study Demonstration*

Fig. 6 illustrates the system behavior under a Q-Learning Controller that jointly adjusts antenna directionality and trans-

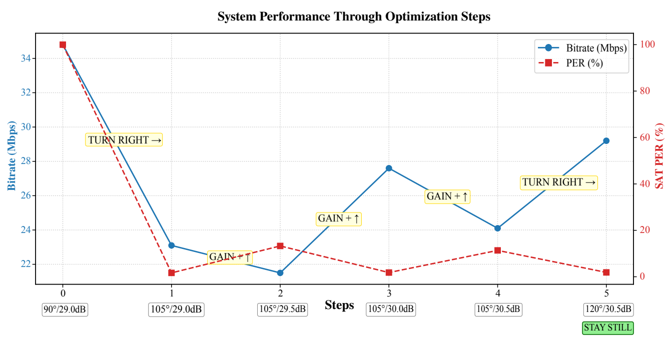

Fig. 6: System Performance through Joint Control

mit gain to optimize SAT performance while minimizing 5G interference. Through Fig. 6, the agent starts from an initial non-optimal input configuration of $90°/29.0$ dB for the Rural Macro case study, aggregating a substantial 5G bit rate at the expense of a high SAT PER. Initially, the agent adjusts the directionality right, observing a 11.7 Mbps bitrate depreciation and 98% PER decrease. Thus, the case study demonstrates how the agent initially prioritizes high directionality control for preservation of the SAT, culminating in a decision that nearly restores ideal SAT performance. However, in response to a restored SAT link, it incrementally increases gain (at a fixed azimuth), improving bitrate of the 5G. After another directional change to $120°$, it achieves a low PER and higher bitrate, converging to a favorable point (annotated by ”stay still” in step 5 of Fig. 6).

Based on this demonstration, the steps exhibit occasional non-monotonic progression: an increase in gain, at times, leads to a decrease in bitrate or a drop in PER, contrary to conventional expectations. These anomalies can be attributed to the introduced Rician Fading and inherent variability of the waveforms; nevertheless, the Q-Learning agent demonstrates from the comprehension of results found in Fig. 4 and Fig. 5 how to make a decision path to balance coexistence and performance, leveraging advantages from adaptive beam and power control.

VI. Conclusion

This work showcased the potential of Joint RRM techniques, by integrating Q-Learning–based power adaptation and directional beam control within an SDR emulation framework. Leveraging the COSMOS testbed for experimentation, Joint RRM significantly preserved DVB-S2 PER performance while simultaneously maintaining the 5G BS link performance at acceptable levels. The algorithm’s reliance on thorough control inputs highlights how spectrum sharing policy not only necessitates data control outside of SAT systems, but also demands algorithms to leverage this data for coexistence scenarios. Moving forward, realistic SDR-based emulation will be required for experimenting with future FR3 implementations; thus, our next experiments will extend Joint RRM framework by incorporating a Transceiver Design that upconverts SDR transmissions from their typical sub-6 GHz range to 12 GHz. This transition will enable coexistence study in real FR3 setting, where both emulation fidelity and regulatory insights can be enhanced. Thus, the design of computational algorithms that account for hardware will demonstrate spectrum sharing policy with more computational optimization, power control, and directionality in an actual FR3 environment. The versatility of applications and models produced through this transceiver experimentation can improve previous considerations and advance future applications of SDR based Emulation of coexistence study between SAT and 5G terrestrial networks.

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